Sine and cosine functions

1 Differentiate the following functions:

a $y = \cos 3x$

b $y = \sin 7x$

c $y = 2 - 3\cos x$

d $y = \sin x + 3\cos x$

e $y = -3\sin x$

f $y = \sin\left(x^6 + 5\right)$

g $y = \sin\left(4x^5 + 3\right)$

h $y = 3x + \sin 5x$

i $y = \sin\left(x^4 + 3\right)$

j $y = \cos\left(3x^2\right)$

k $y = \sin\left(\frac{x}{3}\right) - 2\cos x$

l $y = \sin x^5$

m $y = 4\sin\left(\frac{t}{4}\right) + 3\cos 4t + t^4$

2 Differentiate the following functions:

a $y = 4x\cos x$

b $y = \sin x \cos x$

c $y = \sin^6(3x)$

d $y = \sin 4x\,(2 + \cos x)$

e $y = \sqrt{\cos 4x}$

f $y = \cos^2\left(x + \frac{\pi}{2}\right)$

g $y = \sin^2(5x)$

h $y = \dfrac{\sin x}{x - 8}$

i $y = \left(x^4 + 3\right)\cos 5x$

j $y = \cos 4x \sin 6x$

k $y = x^{\frac{7}{4}}\cos 2x$

l $y = 3\cos 4x + 6\sin 5x$

m $y = \dfrac{3x}{\sin x}$

n $y = x^4 \sin\left(\frac{x}{2}\right)$

o $y = x^3\cos x$

p $y = 4\sin 5x \cos 5x$

Applications

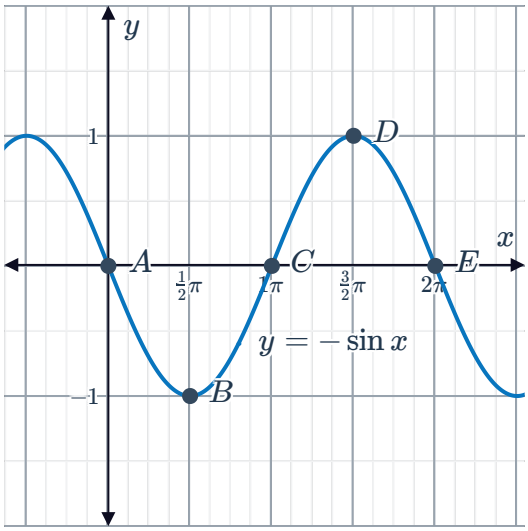
3 Differentiate the function $xy = \sin 5x$.

4 Consider $\tan x = \dfrac{\sin x}{\cos x}$ and differentiate $y = \tan x$ using the quotient rule.

5 Explain why the gradient of the function $y = \sin^2(5x) + \cos^2(5x)$ equal to 0 for all x.

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6 Consider the graph of $y = -\sin x$ which is the gradient function of $y = \cos x$. A number of points have been labelled on the graph.
Name the point on the gradient function that corresponds to the following locations on the graph of $y = \cos x$:



a Where $y = \cos x$ is increasing most rapidly.

b Where $y = \cos x$ is decreasing most rapidly.

c Where $y = \cos x$ is stationary.

7 There is an expansion system in mathematics that allows a function to be written in terms of powers of x. The value of $\sin x$ and $\cos x$, for any value of x, can be given by the expansions below:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \ldots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \ldots$$

Use the expansions to find:

a $\frac{d}{dx}(\sin x)$ in terms of x.

b $\frac{d}{dx}(\cos x)$ in terms of x.

c Hence express the derivatives of $\sin x$ and $\cos x$ in simplest form.

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